SANJIVANI UNIVERSITY, KOPARGAON

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
SPECIALIZATION IN CYBER SECURITY

PRACTICAL ASSIGNMENT REPORT

NPC – UNIT 4: CONFIGURE DHCP AND SET UP A NETWORK USING CISCO PACKET TRACER

Dynamic IPv4 Address Allocation, Pool Scoping, Service Verification & ICMP Connectivity Testing

Student Name	**Vishal Katyare**
Permanent Registration Number (PRN)	**2125UCSM1034**
Course / Degree	Bachelor of Technology (B.Tech)
Department	Computer Science Engineering
Specialization	Cyber Security
Semester / Year	3rd Semester / Second Year (2026–2027)
Subject / Module	Network Protocols and Communication (NPC – Unit 4)
Institution	Sanjivani University, Kopargaon
Repository URL	https://github.com/vishal937/unit-4-npc.git

1. Aim and Objectives

Aim: To design and configure a Local Area Network (LAN) in Cisco Packet Tracer, assign a static IPv4 address to a central server, enable and configure the Dynamic Host Configuration Protocol (DHCP) service, dynamically allocate IP configurations to multiple client PCs, and verify end-to-end communication via ICMP ping.

Objectives:
1. Deploy a network topology comprising 3 client PCs (PC0, PC1, PC2), a Cisco Catalyst 2960 Layer-2 switch (Switch0), and a generic DHCP server (Server0).
2. Assign a static IPv4 address (192.168.1.2/24) to the DHCP server's FastEthernet0 interface.
3. Enable the DHCP service daemon and configure a custom address pool named NPC_POOL.
4. Transition client PCs from static IP configuration to dynamic DHCP allocation.
5. Verify that client PCs automatically lease sequential IP addresses within the designated pool range.
6. Test full bidirectional network connectivity across all client stations and the server using the ICMP ping utility.

2. Hardware & Software Requirements

Component	Device / Software	Role & Configuration Description
Simulation Software	Cisco Packet Tracer v9.0.1	Virtual network modeling and packet simulation environment
DHCP Server	Generic Server-PT (Server0)	Hosts the DHCP daemon; static IP 192.168.1.2/24
Layer-2 Switch	Cisco Catalyst 2960-24TT	Connects nodes into single broadcast domain (VLAN 1)
Client Workstations	3 × Generic PC-PT (PC0, PC1, PC2)	Dynamic DHCP clients receiving leased IP configurations
Physical Cabling	Copper Straight-Through (UTP)	FastEthernet link connections between switch and endpoints

3. Theoretical Principles: The DHCP DORA Process

The Dynamic Host Configuration Protocol (DHCP) operates at the Application Layer (utilizing UDP ports 67 and 68) to automate IP parameter assignment. When a client PC boots or enables DHCP, it undergoes the four-phase **DORA** exchange:

• **Discover (DHCPDISCOVER):** The client broadcasts a layer-2 and layer-3 packet (Dest: 255.255.255.255, FF:FF:FF:FF:FF:FF) searching for available DHCP servers on the local subnet.

• **Offer (DHCPOFFER):** The DHCP server reserves an unassigned IP from its pool and broadcasts an offer containing the proposed IP address, subnet mask, default gateway, and lease duration.

• **Request (DHCPREQUEST):** The client responds with a formal broadcast requesting the offered parameters and notifying other potential DHCP servers that their offers were declined.

• **Acknowledge (DHCPACK):** The server commits the lease to its binding database and sends a unicast/broadcast acknowledgment, allowing the client to initialize its network interface.

4. Network Topology & DHCP Pool Parameters

Configuration Parameter	Configured Value	Operational Function
Server0 IPv4 Address	192.168.1.2	Static management IP address of the DHCP server
Subnet Mask	255.255.255.0 (/24)	Defines the local Class C broadcast network
DHCP Service State	**On**	Enables the background DHCP listening daemon
Pool Name	NPC_POOL	Designated identifier for the IP allocation scope
Start IP Address	192.168.1.10	First assignable dynamic host address in the pool
Maximum Number of Users	50	Restricts pool size to addresses 192.168.1.10 through .59
Default Gateway / DNS	0.0.0.0	Single broadcast segment; external routing not required

5. Step-by-Step Configuration Procedure

Step 1 — Topology Deployment & Cabling:
Place one Cisco 2960 switch (Switch0), one generic server (Server0), and three client PCs (PC0, PC1, PC2) in the Packet Tracer workspace. Connect each device using Copper Straight-Through Ethernet cables: Server0 to Fa0/1, PC0 to Fa0/2, PC1 to Fa0/3, and PC2 to Fa0/4.

Step 2 — Static Server IP Assignment:
Open Server0 ➔ Desktop ➔ IP Configuration. Select the **Static** radio button. Enter IPv4 Address 192.168.1.2 and Subnet Mask 255.255.255.0.

Step 3 — DHCP Service & Pool Configuration:
Navigate to Server0 ➔ Services ➔ DHCP. Turn Service status to **On**. Set Pool Name to NPC_POOL, Start IP Address to 192.168.1.10, Subnet Mask to 255.255.255.0, and Maximum Number of Users to 50. Click **Save** / **Add**.

Step 4 — Client Dynamic Allocation:
Open Desktop ➔ IP Configuration on PC0, PC1, and PC2 sequentially. Switch the configuration from Static to **DHCP**. Verify that Packet Tracer displays the prompt: *'DHCP request successful'*.

Step 5 — End-to-End Connectivity Verification:
Open PC0 Command Prompt and issue ICMP ping requests to Server0 (192.168.1.2), PC1 (192.168.1.11), and PC2 (192.168.1.12). Record transmission metrics and latency.

6. Visual Evidence & Configuration Screenshots

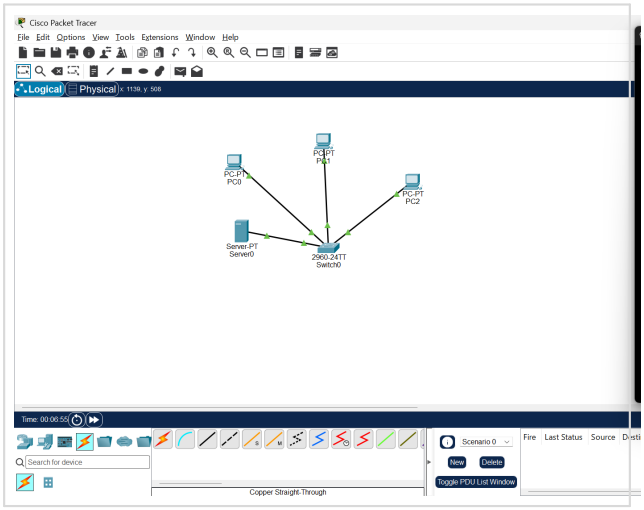


Figure 1: Complete Cisco Packet Tracer Network Topology with Operational Link Lights

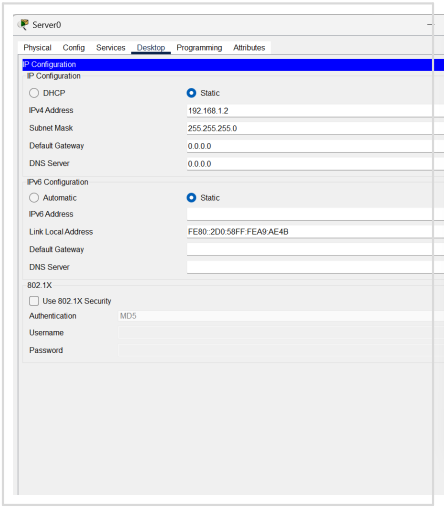


Figure 2: Server0 Static IPv4 Address Configuration (192.168.1.2 / 255.255.255.0)

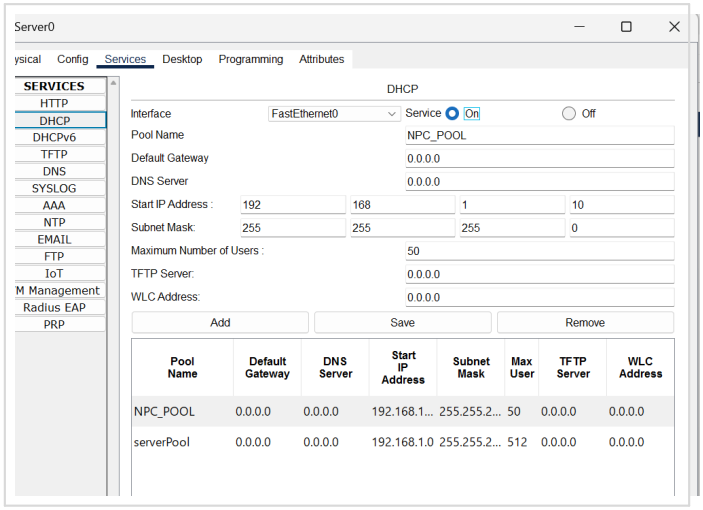


Figure 3: Server0 DHCP Service Daemon Enabled with Custom NPC_POOL Parameters

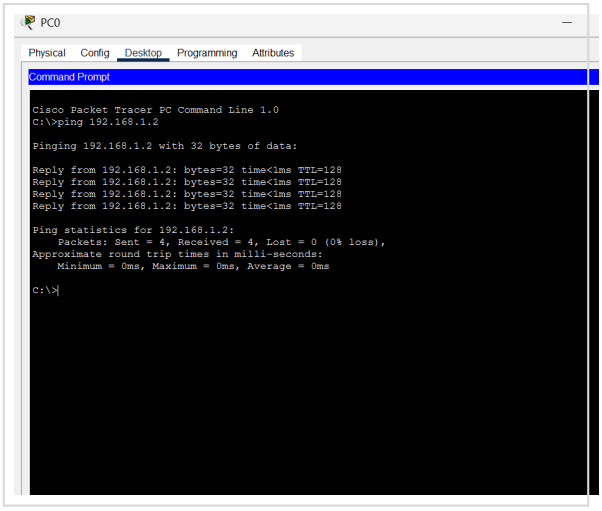


Figure 4: PC0 Dynamic IP Allocation via DHCP (Leased 192.168.1.10 Successfully)

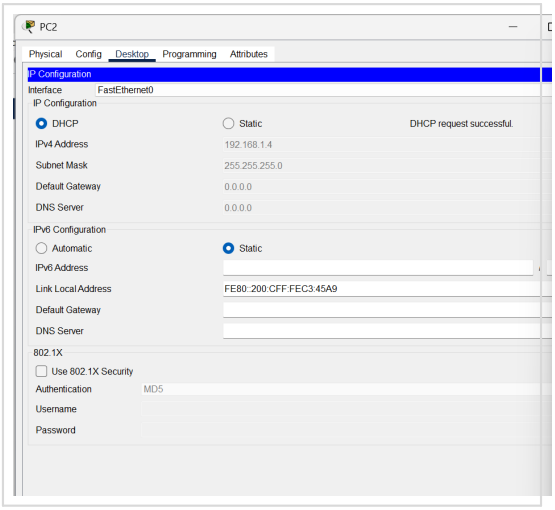


Figure 5: PC1 Dynamic IP Allocation via DHCP (Leased 192.168.1.11 Successfully)

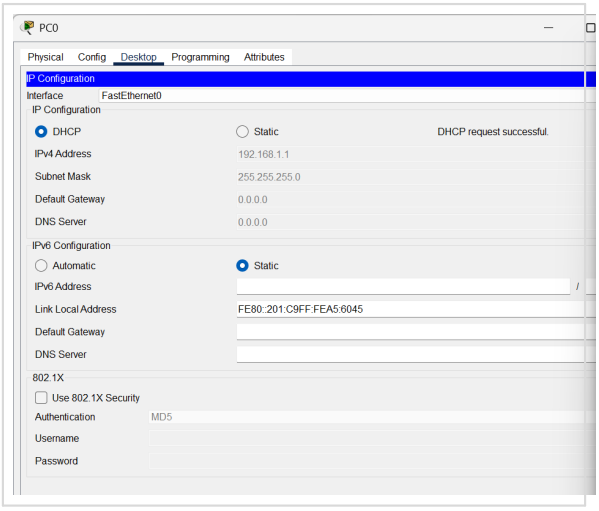


Figure 6: PC2 Dynamic IP Allocation via DHCP (Leased 192.168.1.12 Successfully)

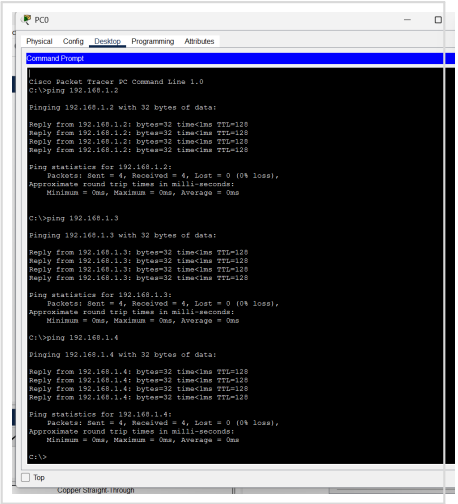


Figure 7: ICMP Ping Connectivity Test from PC0 to Server0, PC1, and PC2 (0% Loss)

7. Results, Empirical Analysis & Conclusion

Client Workstation	Config Mode	Dynamically Leased IP	Subnet Mask	Ping Verification (ICMP)
PC0	DHCP	192.168.1.10	255.255.255.0	**4/4 Received (0% Loss, <1ms)**
PC1	DHCP	192.168.1.11	255.255.255.0	**4/4 Received (0% Loss, <1ms)**
PC2	DHCP	192.168.1.12	255.255.255.0	**4/4 Received (0% Loss, <1ms)**

Conclusion:

The practical assignment successfully demonstrated the automated provisioning of network parameters using Dynamic Host Configuration Protocol (DHCP) inside Cisco Packet Tracer. The central server (Server0) flawlessly leased sequential, non-conflicting IPv4 addresses to all client nodes. Subsequent ICMP ping testing confirmed complete layer-3 connectivity across all subnet stations with zero packet loss and sub-millisecond round-trip times. This experiment confirms that centralized DHCP administration eliminates manual IP assignment errors, prevents address collisions, and simplifies enterprise network scalability.